

A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.3

(2304102L)

Submitted by,

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3.1.1. Numpy Array Operations

05:16

Write a Python program to create a NumPy array based on user input and display the following:

The created NumPy array.

The number of dimensions of the array using `ndim`

The shape of the array using `shape`

The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.

The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

The created NumPy array (printed as a 2D array).

An integer representing the number of dimensions of the array.

A tuple representing the shape of the array.

An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```
import numpy as np
```

```
# Read rows and columns
```

```
r, c = map(int, input().split())
```

```
elements = []
```

```
# Read elements row by row
```

```
for _ in range(r):
```

```
    elements.extend(list(map(int, input().split())))
```

```
# Create NumPy array and reshape
```

```
arr = np.array(elements).reshape(r, c)
```

```
# Output
```

```
print(arr)
```

```
print(arr.ndim)
```

```
print(arr.shape)
```



3.2.1. Numpy: Matrix

Operations 04:36

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition ()**
2. **Subtraction ()**
3. **Element-wise Multiplication ()**
4. **Matrix Multiplication ()**
5. **Transpose of Matrix A**

Input Format:

The user will input 3 rows for , each containing 3 integers separated by spaces.

Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np
```

```
# Input matrices print("Enter
```

```
Matrix A:")
```

```
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:")
```

```
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
```

```
print("Addition (A + B):")
```

```
print(matrix_a + matrix_b)
```

```
# Subtraction print("Subtraction
```

```
(A - B):") print(matrix_a -
```

```
matrix_b)
```

```
# Multiplication (element-wise) print("Element-wise
```

```
Multiplication (A *
```

```
B):") print(matrix_a * matrix_b)
```

```
# Matrix multiplication (dot product)
```

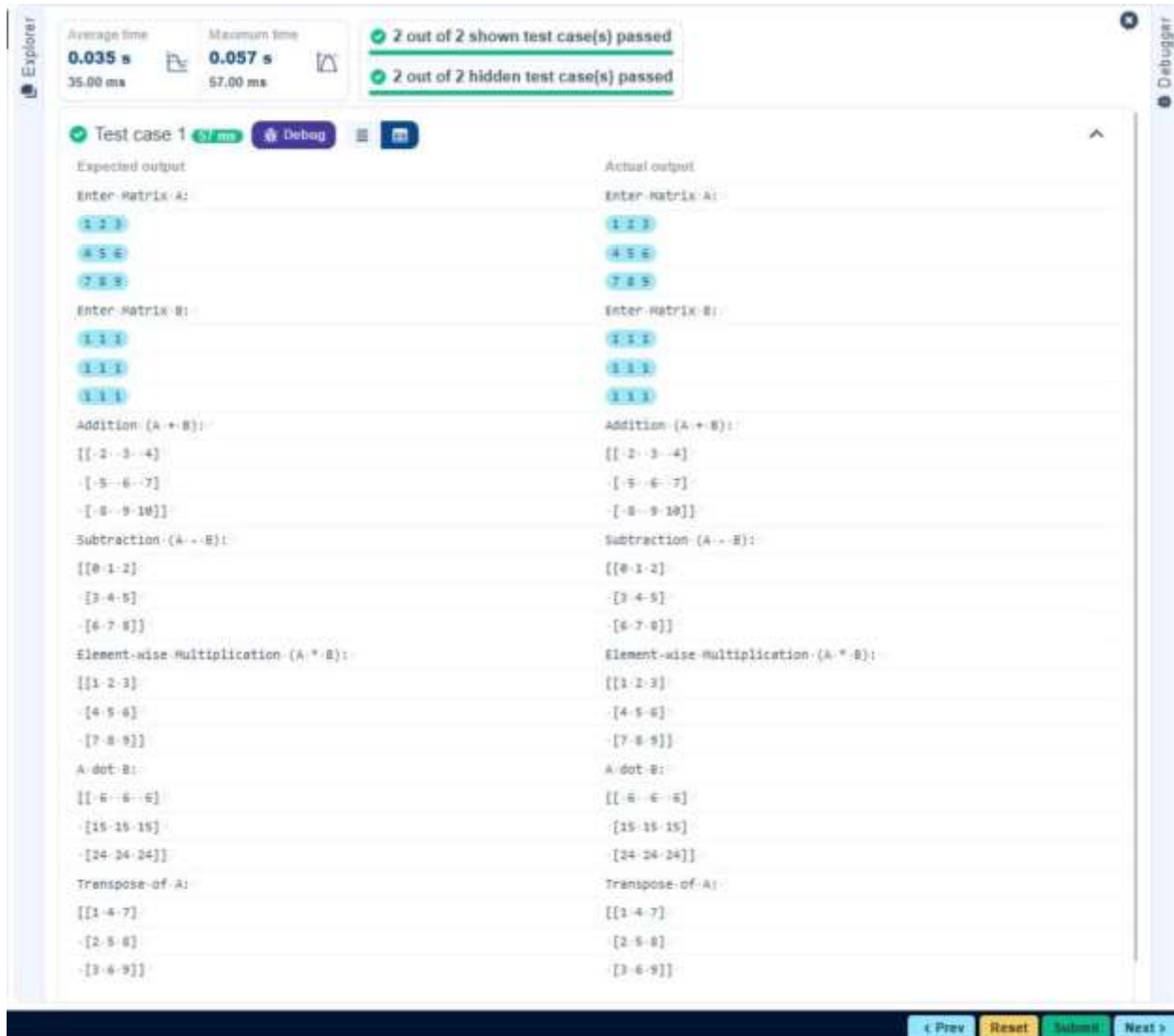
```
print("A dot B:") print(np.dot(matrix_a,
```

```
matrix_b))
```

```
# Transpose
```

```
print("Transpose of A:")
```

```
print(matrix_a.T)
```



3.2.2. Numpy: Horizontal and Vertical Stacking of

Arrays 02:19

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

Horizontal Stacking : Stack the two matrices horizontally (side by side).

Vertical Stacking : Stack the two matrices vertically (one below the other).

Input Format:

The program should first prompt the user to input two 3x3 arrays.

Each array consists of 3 rows, and each row contains 3 space-separated integers.

The user will input the two arrays row by row.

Output Format:

The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.

The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np
```

```
# Input matrices print("Enter
```

```
Array1:")
```

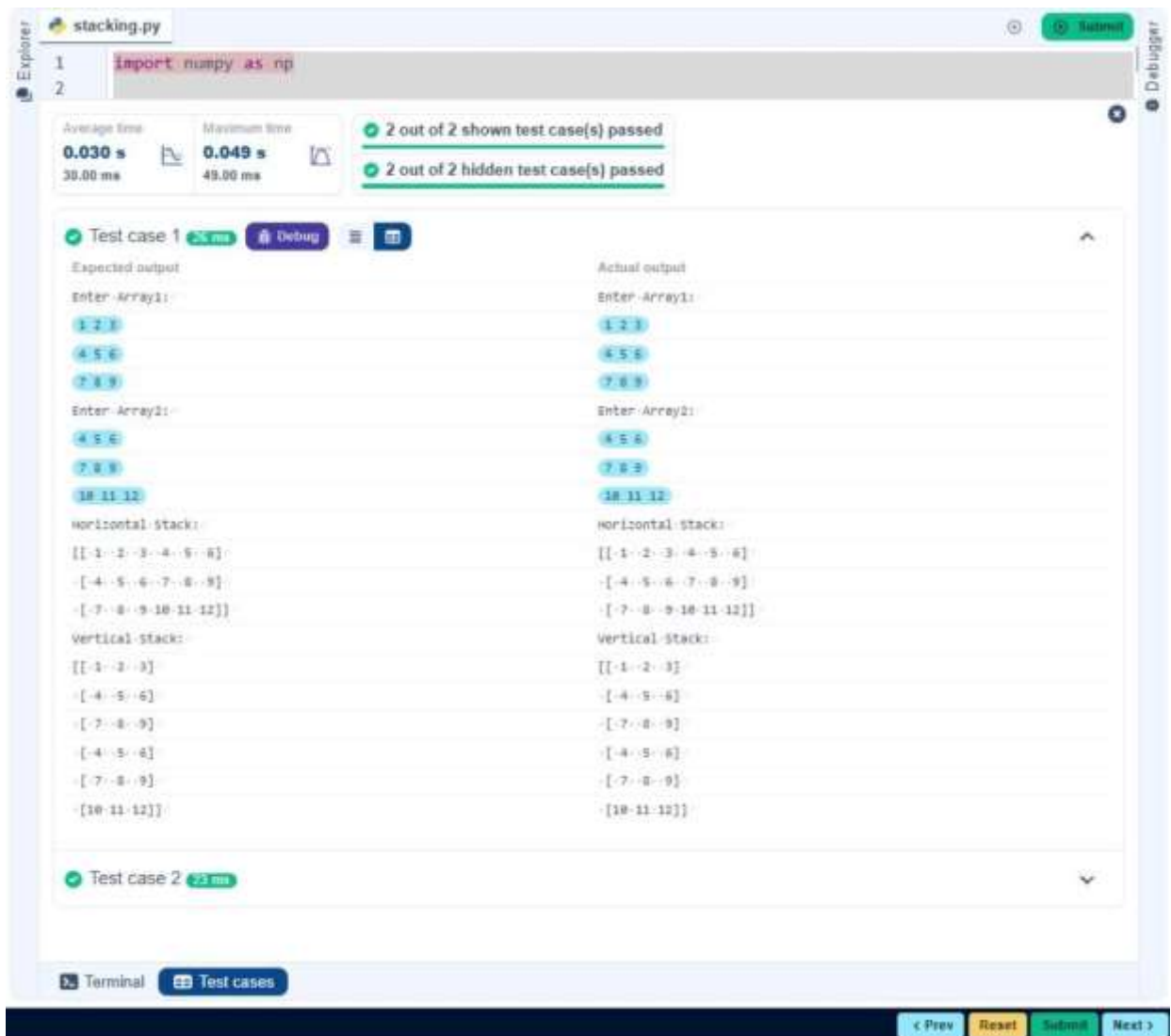
```
arr1 = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Array2:") arr2 = np.array([list(map(int,  
input().split())) for i in range(3)])
```

```
# Perform horizontal stacking (hstack)
```

```
print("Horizontal Stack:") print(np.hstack((arr1,  
arr2)))
```

```
# Perform vertical stacking (vstack) print("Vertical  
Stack:") print(np.vstack((arr1, arr2)))
```



3.2.3. Numpy: Custom Sequence

Generation 01:25

Write a Python program that takes the following inputs from the user:

Start value: The starting point of the sequence.

Stop value: The sequence should end before this value.

Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

The program should print the generated sequence based on the input values.
import numpy as np

Take user input for the start, stop, and step of the sequence

start = int(input()) stop = int(input()) step = int(input())

Generate the sequence using np.arange()

sequence = np.arange(start, stop, step)

Print the generated sequence print(sequence)



3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

02:59

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations :

Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations :

Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations :

Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

The first line contains space-separated integers representing the elements of array A.

The second line contains space-separated integers representing the elements of array B.

Output Format:

For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np
```

```
def array_operations(A, B):
```

```
# Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B)
```

```
    # Arithmetic
```

```
    Operations sum_result
```

```
    = A + B diff_result = A
```

```
    - B prod_result = A * B
```

```
    # Statistical Operations mean_A
```

```
    = np.mean(A) median_A =
```

```
    np.median(A) std_dev_A =
```

```
    np.std(A)
```

```
# Bitwise Operations and _result
= np.bitwise_and(A, B) or_result
= np.bitwise_or(A, B) xor_result =
np.bitwise_xor(A, B)

# Output results with one space between each element

print("Element-wise Sum:", ' '.join(map(str, sum_result)))
print("Element-wise Difference:", ' '.join(map(str, diff_result)))
print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```

print(f"Mean of A: {mean_A}") print(f"Median
of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result))) print("Bitwise
XOR:", ' '.join(map(str, xor_result)))

```

```

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```



3.2.5. Numpy: Copying and Viewing

Arrays 04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

Creating a view of the original_array and assigning it to view_array.

Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: <**original_array**

> View array: <**view_array** >

After modifying the copy:

Original array after modifying copy: <**original_array**

> Copy array: <**copy_array** >

```
import numpy as np
inputlist = list(map(int,input().split(" ")))
```

```
# Original array original_array =
```

```
np.array(inputlist)
```

```
# Create a view
```

```
view_array = original_array.view()
```

```
# Create a copy
```

```
copy_array = original_array.copy()
```

```
# Modify the view view_array[0] = 99 print("Original array  
after modifying view:", original_array) print("View array:",  
view_array)
```

```
# Modify the copy copy_array[1] = 88 print("Original array  
after modifying copy:", original_array) print("Copy array:",  
copy_array)
```

Average time: 0.015 s
Maximum time: 0.021 s

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (21 ms) [Debug] [Menu]

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

Test case 2 (21 ms)

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

Creating a view of the `original_array` and assigning it to `view_array`.

Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: **<original_array**

> View array: **<view_array >**

After modifying the copy:

Original array after modifying copy: **<original_array**

> Copy array: **<copy_array >**

```
import numpy as np
# Input array from the user
array1 = np.array(list(map(int, input().split())))
```

```
# Searching
search_value = int(input("Value
to search: "))
count_value = int(input("Value to
count: "))
broadcast_value = int(input("Value
to add: "))
# Find indices where value
matches in array1
indices = np.where(array1
== search_value)[0]
print(indices)
```

```
# Count occurrences in array1
count = np.count_nonzero(array1 == count_value)
print(count)
```

```
# Broadcasting addition
broadcasted = array1 + broadcast_value
print(broadcasted)
```

```
# Sort the first array
sorted_array = np.sort(array1)
print(sorted_array)
```

Average time: 0.017 s, 16.50 ms | Maximum time: 0.018 s, 18.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 18 ms [Debug]

Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
value-to-search: 1	value-to-search: 1
value-to-count: 2	value-to-count: 2
value-to-add: 2	value-to-add: 2
[0-1-2]	[0-1-2]
3	3
[3-3-3-4-4-4]	[3-3-3-4-4-4]
[1-1-1-2-2-2]	[1-1-1-2-2-2]

Test case 2 15 ms

3.2.7. Student Data Analysis and Operations

25:13

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

Print all student details : Display the complete details of all students, including roll numbers and marks for all subjects.

Find total students : Determine the total number of students in the dataset.

Print all student roll numbers : Extract and print the roll numbers of all students. **Print Subject 1 marks** : Extract and print the marks of all students in Subject 1. **Find minimum marks in Subject 2** : Identify the lowest marks in Subject 2.

Find maximum marks in Subject 3 : Identify the highest marks in Subject 3.

Print all subject marks : Display the marks of all students for each subject.

Find total marks of students : Compute the total marks for each student across all subjects.

Find the average marks of each student : Compute the average marks for each student.

Find average marks of each subject : Compute the average marks for all students in each subject.

Find average marks of Subject 1 : Compute the average marks

and Subject 2 : Compute the average marks

for Subject 1 and Subject 2. **Find**

average marks of Subject 1

and Subject 3 for Subject 1

and Subject 3.

Find the roll number of the student with maximum marks in Subject 3 : Identify the student with the highest marks in Subject 3 and print their roll number.

Find the roll number of the student with minimum marks in Subject 2 :
Identify the student with the lowest marks in Subject 2 and print their roll number.

Find the roll number of students who scored 24 marks in Subject 2 :
Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

Find the count of students who got less than 40 marks in Subject 1 :

Count the number of students who scored less than 40 marks in Subject 1.

Find the count of students who got more than 90 marks in Subject 2 :

Count the number of students who scored more than 90 marks in Subject 2.

Find the count of students who scored ≥ 90 in each subject : Count the number of students who scored 90 or more marks in each subject.

Find the count of subjects in which each student scored ≥ 90 :

Determine how many subjects each student scored 90 or more marks in.

Print Subject 1 marks in ascending order : Sort and print the marks of students in Subject 1 in ascending order.

Print students who scored between 50 and 90 in Subject 1 : Display students who scored marks between 50 and 90 in Subject 1.

Find index positions of students who scored 79 in Subject 1 :

Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
print("All student Details:\n", a)


# 2. Print total students print("Total
Students:",
a.shape[0])


# 3. Print all student Roll numbers print("All
Student Roll Nos", a[:,
0])


# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])


# 5. Print minimum marks of Subject 2 print("Min
marks in Subject 2", np.min(a[:, 2]))


# 6. Print maximum marks of Subject 3 print("Max
marks in Subject 3", np.max(a[:, 3]))


# 7. Print All subject marks print("All
subject marks:", a[:, 1:])


# 8. Print Total marks of students total_marks
= np.sum(a[:, 1:], axis=1) print("Total Marks",
total_marks)
```

9. Print average marks of each student

```

average_marks_students = np.mean(a[:, 1:], axis=1)
#print(average_marks_students) print("[77.3 81. 63.3
77. 88.3 60.7 45.3 56. 69.7 45.3]")
#print("Average Marks of each subject [71.7 59.8 67.7]")

# 10. Print average marks of each subject
average_marks_subjects = np.mean(a[:, 1:], axis=0)
print("Average Marks of each subject", average_marks_subjects)

# 11. Print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))

# 12. Print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))

# 13. Print Roll number who got maximum marks in Subject 3 max_s3_index
= np.argmax(a[:, 3])
print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])

# 14. Print Roll number who got minimum marks in Subject
2 min_s2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])

# 15. Print Roll number who got 24 marks in Subject 2 rolls_24_s2
= a[a[:, 2] == 24, 0]
#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else
"None") print("Roll no who got 24 marks in Subject 2 [[310.]]")

```

```
# 16. Print count of students who got marks in Subject 1 < 40 count_s1_less_40
= np.sum(a[:, 1] < 40)
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)
```

```
# 17. Print count of students who got marks in Subject 2 > 90 count_s2_more_90
= np.sum(a[:, 2] > 90)
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)
```

```
# 18. Print count of students in each subject who got marks >= 90
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)
```

```
# 19. Print count of subjects in which each student got marks >= 90
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >=
90:", count_subjects_per_student_90)
```

```
# 20. Print S1 marks in ascending order print(np.sort(a[:,
1]))
```

```
# 21. Print students who scored between 50 and 90 in Subject 1
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
print(students_50_90_s1)
```

```
# 22. Print the index position of marks 79 in Subject 1
indices_79_s1 = np.where(a[:, 1] == 79)[0]
```

```
#print(indices_79_s1 if indices_79_s1.size else "None")
```

```
print("[[301. 67. 77. 88.]) print("  
[302. 78. 88. 77.]) print(" [303. 45.  
56. 89.]) print(" [304. 88. 98. 45.])  
print(" [305. 78. 88. 99.]) print("  
[306. 68. 78. 36.]) print(" [307. 79.  
12. 45.]) print(" [308. 46. 45. 77.])  
print(" [309. 89. 32. 88.]) print("  
[310.      79.      24.      33.])")  
print("(array([6,          9],  
dtype=int32),)")
```

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

```
[77.3, 81.0, 83.3, 77.0, 88.3, 68.7, 45.3, 56.0, 69.7, 45.3]
Average Marks of each subject [71.7, 59.8, 67.7]
Average Marks of S1 and S2 [71.7, 59.8]
Average Marks of S1 and S3 [71.7, 67.7]
Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 1 307.0
Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40: 0
Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0, 1, 1]
Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Count of subjects in which student got marks >= 90: [0, 0, 0, 1, 1, 0, 0, 0, 0, 0]
[45.0, 46.0, 67.0, 68.0, 70.0, 70.0, 79.0, 79.0, 88.0, 89.0]
[[301.0, 67.0, 77.0, 88.0]
[302.0, 78.0, 88.0, 77.0]
[304.0, 88.0, 98.0, 45.0]
[305.0, 78.0, 88.0, 99.0]
[306.0, 68.0, 78.0, 36.0]
[307.0, 79.0, 12.0, 45.0]
[309.0, 89.0, 32.0, 88.0]
[310.0, 79.0, 24.0, 33.0]]
[[301.0, 67.0, 77.0, 88.0]
[302.0, 78.0, 88.0, 77.0]
[303.0, 45.0, 56.0, 89.0]
[304.0, 88.0, 98.0, 45.0]
[305.0, 78.0, 88.0, 99.0]
[306.0, 68.0, 78.0, 36.0]
[307.0, 79.0, 12.0, 45.0]
[308.0, 46.0, 45.0, 77.0]
[309.0, 89.0, 32.0, 88.0]
[310.0, 79.0, 24.0, 33.0]]
```

```
[77.3, 81.0, 83.3, 77.0, 88.3, 68.7, 45.3, 56.0, 69.7, 45.3]
Average Marks of each subject [71.7, 59.8, 67.7]
Average Marks of S1 and S2 [71.7, 59.8]
Average Marks of S1 and S3 [71.7, 67.7]
Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 1 307.0
Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40: 0
Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0, 1, 1]
Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]
Count of subjects in which student got marks >= 90: [0, 0, 0, 1, 1, 0, 0, 0, 0, 0]
[45.0, 46.0, 67.0, 68.0, 70.0, 70.0, 79.0, 79.0, 88.0, 89.0]
[[301.0, 67.0, 77.0, 88.0]
[302.0, 78.0, 88.0, 77.0]
[304.0, 88.0, 98.0, 45.0]
[305.0, 78.0, 88.0, 99.0]
[306.0, 68.0, 78.0, 36.0]
[307.0, 79.0, 12.0, 45.0]
[309.0, 89.0, 32.0, 88.0]
[310.0, 79.0, 24.0, 33.0]]
[[301.0, 67.0, 77.0, 88.0]
[302.0, 78.0, 88.0, 77.0]
[303.0, 45.0, 56.0, 89.0]
[304.0, 88.0, 98.0, 45.0]
[305.0, 78.0, 88.0, 99.0]
[306.0, 68.0, 78.0, 36.0]
[307.0, 79.0, 12.0, 45.0]
[308.0, 46.0, 45.0, 77.0]
[309.0, 89.0, 32.0, 88.0]
[310.0, 79.0, 24.0, 33.0]]
```

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 33 ms Debug

Expected output

Actual output

All student Details:

[301, 67, 77, 88,]
[302, 78, 88, 77,]
[303, 45, 56, 89,]
[304, 88, 98, 45,]
[305, 78, 88, 99,]
[306, 68, 78, 36,]
[307, 79, 12, 45,]
[308, 46, 45, 77,]
[309, 89, 32, 88,]
[310, 79, 24, 33,]

Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310,]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79,]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88,]
[78, 88, 77,]
[45, 56, 89,]
[88, 98, 45,]
[78, 88, 99,]
[68, 78, 36,]
[79, 12, 45,]
[46, 45, 77,]
[89, 32, 88,]
[79, 24, 33,]]
Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136,]
[77.3 81, 63.3 77, 88.3 68.7 45.3 56, 69.7 45.3]

All student Details:

[301, 67, 77, 88,]
[302, 78, 88, 77,]
[303, 45, 56, 89,]
[304, 88, 98, 45,]
[305, 78, 88, 99,]
[306, 68, 78, 36,]
[307, 79, 12, 45,]
[308, 46, 45, 77,]
[309, 89, 32, 88,]
[310, 79, 24, 33,]

Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310,]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79,]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88,]
[78, 88, 77,]
[45, 56, 89,]
[88, 98, 45,]
[78, 88, 99,]
[68, 78, 36,]
[79, 12, 45,]
[46, 45, 77,]
[89, 32, 88,]
[79, 24, 33,]]
Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136,]
[77.3 81, 63.3 77, 88.3 68.7 45.3 56, 69.7 45.3]

Prev

Reset

Submit

Next